

AI-powered carbon accounting & compliance assistant for India's MSMEs

Photograph a bill. Know your emissions. Keep your buyers.

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Ideas of India – Season 2 · Sustainable & Resilient India · Climate Tech &
Carbon Management
Concept-to-Prototype Execution Plan

The problem arrived with two deadlines

31 July 2026 — India's first CCTS carbon-compliance filings fall due. ~490 plants in 7 sectors already carry binding targets; penalty is 2× the carbon-credit price.

1 January 2026 — EU's CBAM entered its paid phase. €8–9 billion of Indian exports now carry a carbon cost (GTRI: 15–22% effective price pressure on coal-based steel & aluminium).

8.7 crore MSMEs — 31.1% of GDP, 48.6% of exports, 38.9 crore jobs — and almost none of them can measure their own carbon footprint. Their buyers are starting to ask.

Who suffers: the owner with no sustainability team

**A tile maker in Morbi. A dyeing unit owner in
Tiruppur.**

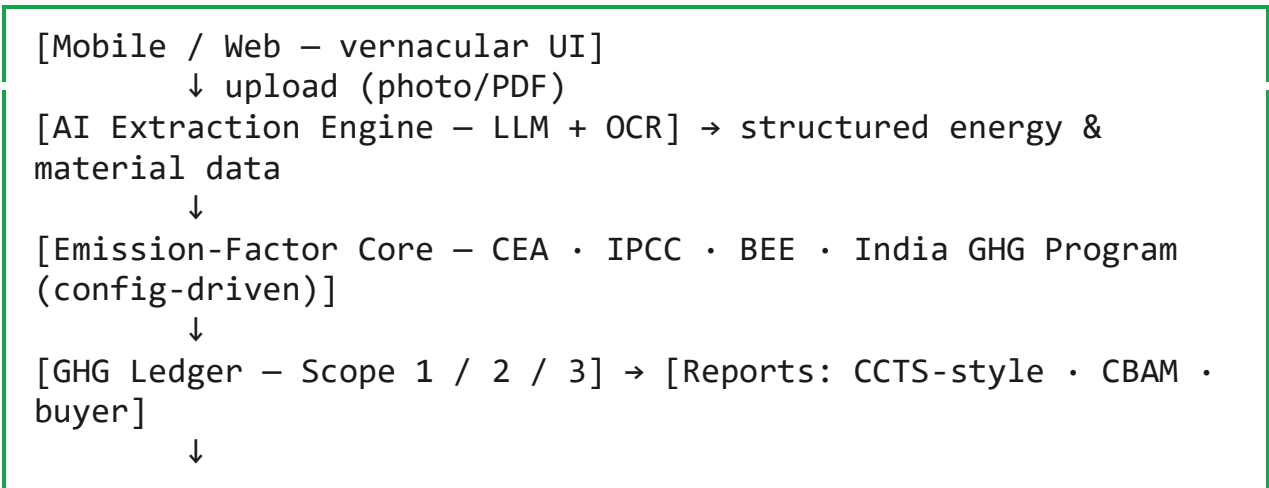
- Records = paper bills + photos on a phone + WhatsApp
- Consultants cost more than his yearly paperwork budget; enterprise carbon software is English-only, corporate-priced, months to deploy
- His buyer just asked for emissions data. He has no way to answer.

The gap: no tool combines AI document reading + Indian languages + India-specific factors + compliance outputs + an MSME price. That empty quadrant is CarbonSathi.

The solution: 4 steps, minutes not weeks

Step	What happens
1. Upload	Owner photographs electricity bills / fuel invoices — any format, any language
2. AI reads	LLM + OCR extraction turns documents into structured consumption data
3. Compute	India-native factor engine (CEA grid factor 0.7117 tCO ₂ /MWh v21.0, IPCC, India GHG Program) → Scope 1, 2, 3 emissions
4. Act	Live dashboard · CCTS-Form-A-style & CBAM/buyer-ready reports · savings actions ranked by payback · water & material-waste tracking · credit-readiness score

Architecture (built on a proven stack)



[Recommendation Engine – savings ranked by payback · credit-readiness score]

Stack: React · Node.js/Express · MongoDB · Redis/BullMQ queues · Socket.IO real-time · JWT/OAuth · Docker/CI-CD. Stateless and queue-based, so it scales horizontally.

Status: functional prototype, honestly stated

Working today

- End-to-end flow runs: upload → AI extraction → CO₂e calculation → dashboard → PDF report mirroring CCTS Form-A fields
- Factor outputs cross-checked against published CEA/BEE values; tested on simulated datasets for two unit archetypes (ceramic kiln, textile dyeing)

Not yet done

- Live commercial pilot · verifier (ACVA) integration · full language coverage

Execution credibility: applicant has shipped two production platforms on this stack — a real-time collaborative coding/interview platform with AI evaluation, and a fault-tolerant webhook security service (zero payload loss under concurrent load).

Why nothing else fills this gap

	Enterprise (Persefoni, Watershed, Sweep)	Indian ESG tools (sentra.world, Greenio, Zoei)	IIM-B free tool	Consultants	CarbonSathi
MSME price	No	Partly	Free	No	₹500– 2,000/mo

AI reads documents	No	No	No (manual entry)	Human	Yes
Indian languages	No	No	No	Sometimes	10+ planned
CCTS/CBAM outputs	No	Partial	No	Yes, costly	Yes, native
Serves lakhs of units	No	No	No	No	Yes — software scales

Result: ~100× cheaper than the consultant route; measurement drops from weeks to minutes.

Year-1 pilot targets (conservative, sourced)

Outcome	Target	Basis
Units onboarded	200 (Morbi + Tiruppur)	2 best-documented BEE SME clusters
CO ₂ measured / reduced	~35,000 / ~2,500 tCO ₂ e per year	CEA factor 0.7117 tCO ₂ /MWh; BEE SME audit savings rates
Energy saved	~3.5 GWh/yr (5–15% per unit)	BEE SME cluster audit findings
Solar enabled	~2.2 GWh/yr (~20 units × 75 kW)	BEE facilitated 800 kWp in 18 Thangadh ceramic plants; 2.8–3.4-yr payback
Water conserved	100–150 million litres/yr	Dyeing ~200 L/kg fabric; Tiruppur halved water use in a decade (CII)
Cost savings	₹4 crore+ aggregate (~₹2 lakh/unit)	Energy spend benchmarks, BEE audits
Jobs	25–30 "Carbon Mitra" roles	Cluster field-support model

18-month roadmap & budget (₹12 lakh pilot)

Phase	Months	Milestones
Done	—	Functional prototype: AI pipeline, factor engine, dashboard, reports
1. Field-ready	1–3	Cluster association LOIs (Morbi, Tiruppur) · top-3 language UI · methodology note published
2. Pilot	4–9	90-day pilot with 15–20 units → scale to 200 · field accuracy tuning · first paid tiers
3. Prove & integrate	10–14	Verified savings case studies · ACVA/verifier integration path · incubator + RAMP linkage
4. Scale	15–18	2 more clusters · association licensing model · 10+ languages

Budget: cloud & AI ₹2.5L · localization ₹1.5L · field onboarding & training ₹4L · travel ₹1.5L · methodology/verification advisory ₹1.5L · contingency ₹1L

Why this wins, and the ask

Regulation has made carbon measurement mandatory. CarbonSathi makes it possible for the 8.7 crore businesses that were left out.

- **Timely:** rides two live deadlines (CCTS 31 July 2026, CBAM paid phase)
- **Real:** prototype runs end to end today, built by someone who has shipped production AI platforms
- **Aligned:** NDC 45% intensity cut · Net Zero 2070 · Mission LiFE · Viksit Bharat 2047 · SDGs 7/8/9/12/13
- **Scalable:** software + cluster-association distribution; CCTS pull toward 740+ obligated entities

The ask: jury validation and mentorship to convert the pilot pipeline into signed cluster LOIs — and take India's carbon market to the base of its economy.

**India cannot manage what six crore of its businesses cannot measure.
CarbonSathi makes them count.**